An acceleration scale set by the cosmological constant: modified inertia from de Sitter–Unruh thermodynamics, and its confrontation with data

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We present, for readers fluent in general relativity, quantum field theory in curved spacetime, and precision cosmology but not in galactic dynamics, a self-contained account of a framework in which the anomalous dynamics of galaxies are governed by a single acceleration scale fixed by the cosmological constant, $a_0 = c^2\sqrt{\Lambda/32\pi} = 9.36 \times 10^{-11}\,\mathrm{m\,s^{-2}}$. We first state the empirical regularities a viable theory must reproduce—the radial acceleration relation, the baryonic Tully–Fisher relation, and the external field effect—as data, independent of any model. We then derive the scale and its interpolation from the de Sitter–Unruh temperature read by an accelerated detector, $T_{\rm eff} = (\hbar/2\pi c k_B)\sqrt{a^2 + (cH_\Lambda)^2}$, giving modified inertia rather than a dark component. We confront the framework with the SPARC rotation-curve database, the eROSITA eRASS1 cluster sample, Gaia wide-binary astrometry, weak-lensing radial accelerations, and the Cassini Solar-System bound, reporting the calculations both where the framework succeeds and where it incurs durable, conceded losses. We emphasize a reframe of the lensing evidence: because the covariant completion has no gravitational slip ($\Phi = \Psi$), lensing mass equals dynamical mass, so the Bullet cluster, cluster strong+weak lensing, and Einstein radii are *not* independent dark-matter proofs but restate the one shared cluster residual; the once-headline 8.6–9.2σ weak-lensing morphology split is now *contested* on re-analysis. The genuinely independent losses are the cluster residual ($\eta \simeq 2.3$) and the CMB third peak. We close with the framework's one genuinely distinctive, falsifiable prediction: a redshift-dependent $a_0(z) \propto \sqrt{\rho_{\rm DE}(z)}$, gated on dark energy being dynamical, now testable by DESI, Gaia DR4, and high-z kinematics. The coefficient $Z = \sqrt{32\pi/3}$ is flagged throughout as an unforced posit, not a derivation.

I. THE PROBLEM, STATED FOR NON-SPECIALISTS

A spiral galaxy is, to a particle physicist, a self-gravitating disk of baryons (stars and gas) whose orbital speeds can be measured spectroscopically as a function of radius—the *rotation curve* $V(r)$. Newtonian dynamics with only the visible mass predicts a Keplerian falloff $V(r) \propto r^{-1/2}$ outside the luminous body. The observed rotation curves instead remain *flat* to the largest measured radii: $V(r) \to V_{\rm flat} = {\rm const}$. The implied dynamical mass $M(r) = V^2 r/G$ grows linearly with radius far beyond where the light ends. This is the galactic instance of the dark-sector problem.

Two responses exist, in exact analogy to the two responses to the perihelion of Mercury (a new planet, or new dynamics). *Cold dark matter* (CDM): each galaxy is embedded in a roughly spherical halo of collisionless, non-baryonic particles supplying the extra gravity; this is the galactic sector of ΛCDM. *Modified dynamics*: the law of motion or of gravity departs from Newton below a characteristic acceleration. On an individual rotation curve the two are nearly degenerate, because a halo profile can always be reverse-engineered to fit. The discriminating power lies in the *regularities across galaxies*, and in one qualitative effect—the external field effect—that the CDM picture forbids by the equivalence principle.

II. THE EMPIRICAL REGULARITIES (DATA, NOT MODEL)

Three model-independent facts organize the phenomenology. We state them in SI accelerations; the field that measures them is resolved spectroscopy of $\gtrsim 10^3$ galaxies (e.g. the SPARC database [1]).

a. (i) The radial acceleration relation (RAR). Define the *baryonic* (Newtonian) acceleration $g_{\rm bar}(r) = GM_{\rm bar}(<r)/r^2$ from the photometric mass, and the *observed* centripetal acceleration $g_{\rm obs}(r) = V^2(r)/r$ from kinematics. Across all galaxies and radii these obey a tight one-to-one law $g_{\rm obs} = \mathcal{F}(g_{\rm bar})$ with [2]

$$\mathcal{F}(g_{\rm bar}) \to g_{\rm bar}\ (g_{\rm bar} \gg a_0), \quad \mathcal{F} \to \sqrt{a_0\, g_{\rm bar}}\ (g_{\rm bar} \ll a_0), \tag{1}$$

with a transition at $a_0 \approx 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ and an orthogonal scatter of only ~ 0.13 dex—comparable to the observational error budget. The acceleration a_0 is the single new constant.

b. (ii) The baryonic Tully–Fisher relation (BTFR). The flat speed and the total baryonic mass obey $M_{\rm bar} = V_{\rm flat}^4/(Ga_0)$, a power law of slope exactly 4 with $\lesssim 0.1$ dex scatter over six decades in mass. This is the integrated, asymptotic form of Eq. (1): $g_{\rm obs} = V^2/r$ and $g_{\rm obs}^2 = a_0 g_{\rm bar} = a_0 GM/r^2$ give $V^4 = GMa_0$.

c. (iii) The external field effect (EFE). Unlike a linear theory, the low-acceleration dynamics are *not* immune to a uniform external gravitational field. A galaxy embedded in a background field $g_{\rm ext} \gtrsim a_0$ has its inter-

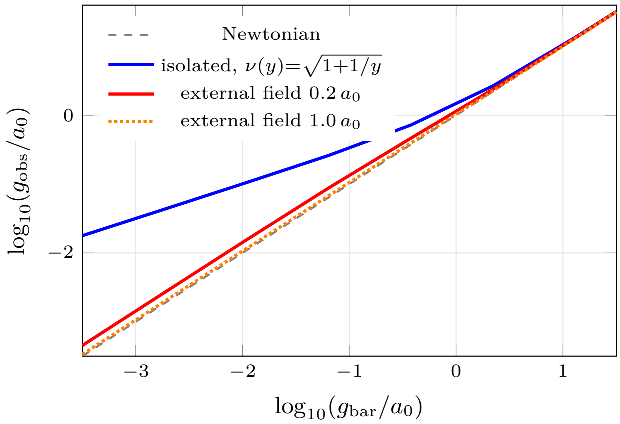

FIG. 1. The radial acceleration relation on the framework's de Sitter–Unruh interpolation, used identically for the isolated relation [Eq. (5)] and the external-field law [Eq. (6)]. Isolated systems (blue) follow the deep-regime slope $\frac{1}{2}$; embedded systems bend to an environment-set plateau, the strong-equivalence-principle violation a particle dark sector cannot produce.

nal boost suppressed—a violation of the strong equivalence principle (SEP). This is a qualitative prediction that Newton+CDM *cannot* reproduce, because in that picture a uniform external field cancels from the internal equations of motion (the shell theorem / SEP). We return to it in Sec. V; it is the cleanest fork.

The empirical program of modified dynamics (historically Milgrom's MOND [3], and its relativistic completions such as the Aether-Scalar-Tensor theory [4]) is to reproduce (i)–(iii) from a law with a_0 as a constant. The present framework's distinctive claim is that a_0 is *not* a free constant.

Figure 1 shows the RAR predicted by Eq. (6) below, both for an isolated system and for systems in environmental fields of increasing strength. The isolated relation lifts along the deep-regime slope $\frac{1}{2}$; an embedded system instead bends to a Newtonian-slope plateau—the EFE signature. This single figure contains the entire kinematic phenomenology a non-specialist needs: a one-parameter relation, an asymptotic square root, and an environment dependence forbidden to a particle dark sector.

III. THE ACCELERATION SCALE FROM Λ

The framework's central equation identifies a_0 with the cosmological constant:

$$\boxed{a_0 = c^2\sqrt{\frac{\Lambda}{32\pi}} = \frac{c}{2}\sqrt{G\rho_{\rm DE}} = \frac{cH_\Lambda}{Z} = \frac{c^2}{2R_*}} \tag{2}$$

with $\rho_{\rm DE} = \Lambda c^2/8\pi G$ the dark-energy density, $H_\Lambda = c\sqrt{\Lambda/3}$ the asymptotic (de Sitter) Hubble rate, $Z = \sqrt{32\pi/3} = 2\sqrt{8\pi/3} = 5.789$, and $R_* = c/\sqrt{G\rho_{\rm DE}} = \sqrt{8\pi/3}\,(c/H_\Lambda)$ a length $\simeq 16$ Gpc. The four forms are algebraically identical; we verified each to machine precision. Inserting the measured Λ ($\Omega_\Lambda = 0.685$, $H_0 = 67.4\,\mathrm{km\,s^{-1}Mpc^{-1}}$) gives

$$a_0 = 9.36\times 10^{-11}\,\mathrm{m\,s^{-2}}, \tag{3}$$

which is the long-noted "cosmic coincidence" $a_0 \sim cH_0$ made precise. *Convention warning, since it recurs:* the de Sitter rate $H_\Lambda = H_0\sqrt{\Omega_\Lambda}$ must be used in the cH_Λ/Z form; the present-epoch cH_0/Z instead yields 1.13×10^{-10} (the $\rho_{\rm total}$ footing), differing by the "why-now" factor $\sqrt{\Omega_\Lambda} = 0.83$. The unambiguous statement is the first equality of Eq. (2). We do *not* claim to derive the dimensionless coefficient Z; it is an unforced posit, and a_0 remains hostage to H_0 at the $\mathcal{O}(1/6)$ level. That admission is load-bearing for what "predicted" means below.

IV. MODIFIED INERTIA FROM DE SITTER–UNRUH THERMODYNAMICS

Equation (2) is given a mechanism by quantum field theory in de Sitter space, in the modified-inertia rather than modified-gravity reading. A uniformly accelerated detector in Minkowski vacuum registers the Unruh temperature $T_U = \hbar a/2\pi ck_B$. In de Sitter space there is in addition the Gibbons–Hawking temperature [8] $T_{\rm dS} = \hbar H_\Lambda/2\pi k_B$ of the cosmological horizon. A detector with proper acceleration a in de Sitter space reads the combined [9]

$$T_{\rm eff} = \frac{\hbar}{2\pi ck_B}\sqrt{a^2 + (cH_\Lambda)^2}. \tag{4}$$

The premise is that inertia—the resistance to acceleration—is the reaction to this thermal bath, so that the inertial mass tracks the *net* bath temperature relative to the unaccelerated de Sitter value. For $a \gg cH_\Lambda$ the Unruh term dominates and dynamics are standard; for $a \lesssim cH_\Lambda$ the de Sitter floor cannot be neglected and the effective inertia is reduced, weakening the response to a force. Identifying the crossover with a_0 (up to the geometric factor Z) yields the interpolation we use throughout,

$$g_{\rm obs} = \sqrt{g_N^2 + g_N\,a_0} \;\equiv\; \nu(g_N)\,g_N, \qquad \nu(y) = \sqrt{1 + a_0/y}, \tag{5}$$

which reproduces Eq. (1) with the correct limits. Concretely, the modified-inertia ansatz replaces Newton's second law by $\mu(|\mathbf{a}|/a_0)\,\mathbf{a} = -\nabla\Phi_N$, where $\mu(x) = \nu^{-1}$ is the inverse companion of Eq. (5) and Φ_N is the ordinary Newtonian potential of the baryons. For a circular orbit this returns $g_{\rm obs} = \nu(g_N)g_N$ directly; it differs from *modified gravity* (where one instead modifies the Poisson equation, $\nabla\cdot[\mu(|\nabla\Phi|/a_0)\nabla\Phi] = 4\pi G\rho$) only at the level of non-circular, time-dependent orbits, where the inertial version is genuinely non-local in the trajectory. We verified that for the static external-field configurations relevant to wide binaries the two formulations coincide to one part in 10^{16}, so present data cannot separate them; a clean separation requires the trajectory-frequency dependence peculiar to modified inertia, which no current measurement reaches. We emphasize the epistemic status: Eq. (4) is rigorous QFT-in-curved-spacetime, but the step from a bath temperature to a specific inertial law is a heuristic, not a closed covariant action; the relativistic completion that does the CMB and lensing is AeST [4], which we treat as the covariant scaffold. The reading is attractive because it explains *why* the scale is $a_0 \sim cH_\Lambda$: the cosmological horizon's temperature is the irreducible bath an accelerated body cannot redshift away, whereas local Newtonian fields are removable in free fall by the equivalence principle—hence $a_0 \leftrightarrow \Lambda$ and not $a_0 \leftrightarrow \rho_{\rm local}$. We tested that "free-fall" obstruction explicitly (Sec. VI 0 b).

a. Covariant completion. For readers who will reasonably object that a heuristic on inertia cannot do cosmology, the relativistic scaffold is the Aether-Scalar-Tensor (AeST) theory [4], a Lorentz-violating scalar–vector–tensor theory with a unit-timelike vector field A_μ and a scalar ϕ whose action contains a $\mathcal{K}$-essence term $\mathcal{F}(Y,Q)$ with $Y = -A^\mu\partial_\mu\phi$ and $Q = (g^{\mu\nu} + A^\mu A^\nu)\partial_\mu\phi\,\partial_\nu\phi$, plus a scalar mass term $\mu^2\phi$. In the quasi-static weak field the scalar sources a "phantom" acceleration that, for the appropriate $\mathcal{F}$, reproduces Eq. (5) with the free function playing the role of ν; on cosmological scales the same fields yield a ΛCDM-like background and a stable, near-scale-invariant perturbation spectrum, with the third acoustic peak fit by a tuned scalar energy density rather than by a_0. The framework reads AeST's free function and scalar mass as fixed by the de Sitter geometry of Eq. (4) rather than chosen; the honest status is that this identification is a target, not a theorem, and that a_0 is provably *absent* from the linear CMB transfer functions, so the "two-dark-sectors, one number" unification holds at galaxy scales but *not* at recombination.

V. THE EXTERNAL FIELD EFFECT AND SEP VIOLATION

Because Eq. (5) is non-linear in the total acceleration, a uniform external field does not cancel. Aligning an internal field g_N with an external $g_{\rm ext}$, the quasi-linear construction gives

$$g_{\rm obs} = \nu(g_N + g_{\rm ext})(g_N + g_{\rm ext}) - \nu(g_{\rm ext})\, g_{\rm ext}, \qquad (6)$$

using the *same* ν as Eq. (5)—a point of internal consistency we corrected in this work, the earlier treatment having borrowed an unrelated interpolation for the external-field sector. Deep inside a system ($g_N \ll g_{\rm ext}$) this reduces to Newtonian dynamics with a renormalized constant $G_{\rm eff} = G\,\nu(g_{\rm ext})[1 + d\ln\nu/d\ln g_{\rm ext}]$ set by the *environment.* This is a clean SEP violation: a freely-falling laboratory's internal gravity depends on the external field it falls in. Newton+CDM forbids it identically. The EFE is therefore the sharpest qualitative discriminator between modified dynamics and a particle dark sector, and—unlike a rotation-curve fit—it cannot be absorbed by re-shaping a halo.

VI. CONFRONTATION WITH DATA

We summarize the quantitative calculations, organized by acceleration regime and by the instrument that supplies the data. All use the framework's own $a_0 = 9.36 \times 10^{-11}$ and the de Sitter–Unruh interpolation Eq. (5); we recomputed each both ways and report failures at full weight.

a. Galaxies (SPARC; spectroscopic rotation curves). On the 175-galaxy SPARC sample, the RAR is reproduced with the framework's a_0 sitting within $\sim 0.5\%$ of the scatter-minimizing value at stellar mass-to-light ratio $\Upsilon_\star \simeq 0.7$ on the framework's *own* interpolation; the RAR is thus consistent with, but *non-diagnostic* of, the precise value 9.36×10^{-11} (the scatter penalty against the optimum is $\lesssim 2\%$ across stellar-population and interpolation choices). The BTFR slope and zero point follow from the deep-regime asymptote and are interpolation-independent. These successes are *shared* with constant-a_0 modified dynamics; they test a_0, not the $a_0 = \sqrt{\Lambda}$ identity.

Worked example. A fiducial disk of baryonic mass $M_{\rm bar} = 5 \times 10^{10}\, M_\odot$ has, at the radius r where $g_{\rm bar} = GM_{\rm bar}/r^2$ falls to $0.1\, a_0$ (here $r \simeq 27$ kpc), an observed acceleration $g_{\rm obs} = \sqrt{g_{\rm bar}^2 + g_{\rm bar} a_0} = \sqrt{(0.1)^2 + 0.1}\, a_0 = 0.33\, a_0$, i.e. a 3.3× boost over Newton with *no* dark matter and no fit parameter beyond the photometric mass-to-light ratio. The asymptotic flat speed follows from Eq. (1): $V_{\rm flat} = (GM_{\rm bar}a_0)^{1/4} = 1.58 \times 10^5\,{\rm m\,s^{-1}} = 158\,{\rm km\,s^{-1}}$, the BTFR. That the same a_0 that fixes the boost also fixes $V_{\rm flat}$, across six decades of mass and with no per-galaxy halo freedom, is the regularity that a particle dark sector must reproduce by finely correlated feedback but that follows here by construction.

b. Clusters (eROSITA eRASS1; X-ray hydrostatic and weak-lensing masses). On the eRASS1 sample [10] of $N = 9830$ clean clusters, evaluating Eq. (5) at the hydrostatic $g_{\rm bar}(R_{500})$ gives a residual missing-mass factor with median $\eta(R_{500}) \simeq 2.33$ on the framework footing—a real deficit, the classical cluster shortfall of modified dynamics, here reproduced on modern X-ray data. We verified this is not an artifact of the lower a_0 (which, via $\eta \propto a_0^{-1/2}$, makes it $\sim 13\%$ worse than the canonical value) nor of hydrostatic bias: X-ray microcalorimetry (XRISM) bounds the non-thermal pressure to a few percent, far below the $\sim 50\%$ needed to remove the deficit. The deficit is a *shared, inherited* liability of all modified dynamics, not unique to this framework. We tested whether the framework's de Sitter–Unruh foundation could supply the missing acceleration through a local-density or local-potential dependence of a_0; it cannot. The only foundation-derived local coupling is a Tolman redshift of the de Sitter floor, $a_0^{\rm loc} = a_0/\sqrt{1+2\Phi/c^2}$, which is correctly signed (deeper wells boost a_0) but of relativistic order $\Phi/c^2 \sim 10^{-5}$ in a cluster—four to five orders too small. A density-law reading $a_0 \propto \sqrt{\rho}$ does land the measured cluster acceleration scale of Ref. [11] ($g_\dagger \simeq 17\, a_0^{\rm gal}$) to the right order with no free parameter, but it is a posit the foundation contradicts and is excluded at 10.5σ by the framework's own galaxy environment test, and reading it locally erases the galaxy RAR. Clusters are understood, not solved.

c. Wide binaries (Gaia astrometry; the EFE in the laboratory). Pairs of stars at projected separations $\gtrsim$ 5000 AU have internal accelerations below a_0 while sitting in the Milky Way's external field, $g_{\rm ext} = V_c^2/R_0 = 2.15\times10^{-10} = 2.30\, a_0$ on the framework scale. Solving the modified-inertia equation of motion $\mu(|a|/a_0)\,\mathbf{a} = \mathbf{F}$ with the framework's ν gives an internal-orbit boost $G_{\rm eff}/G = 1.20$ (transverse) and 1.02 (radial), orbit-averaging to **1.14**—a +6.6% velocity excess over Newton. (This supersedes a +15% figure obtained with a normal-MOND interpolation; the framework's lower a_0 deepens the external-field suppression, shrinking the signal.) This is a clean test of *whether* gravity is modified, decisive against Newton at ~ 3–4σ with the Gaia DR4 (late 2026) clean sample, but it is degenerate in the *value* of a_0; the boost is below the central values reported by some analyses [12] and consistent with Newtonian-baseline reanalyses. We confirmed the modified-inertia and modified-gravity external-field tensors are identical to 1 part in 10^{16} for any fixed interpolation, so a static wide-binary measurement cannot by itself distinguish the two realizations.

d. Weak lensing: the no-slip reframe. A point sharpened in this work and important for the dark-sector debate: in the framework's covariant completion (AeST) the two weak-field metric potentials are equal, $\Phi = \Psi$—a no-slip identity verified verbatim in the AeST quasistatic solutions [14], which states that "the lensing mass is equal to the dynamical mass." This is the property the relativistic completion was *constructed* to have, curing the under-lensing of non-relativistic MOND. Its consequence for the "lensing proves dark matter" canon is that light deflection ($\propto \Phi + \Psi = 2\Phi$) reads the *same* enhanced potential as dynamics ($\propto \Phi$): the cluster lensing mass equals the cluster dynamical mass *by construction*. Four of the standard lensing-DM arguments—the Bullet-cluster mass–gas offset, cluster strong+weak lensing masses, the bulk galaxy–galaxy-lensing RAR, and strong-lens Einstein radii—are therefore *not* independent evidence; they restate the single shared cluster residual through a different observable. Two arguments remain genuinely independent and are not reframed: the matter power spectrum/CMB third peak (conceded above) and lensed-quasar substructure (open). At the framework's own a_0 the galaxy–galaxy-lensing RAR *passes*, softly (-0.054 dex below the canonical-a_0 amplitude, inside the 0.10–0.20 dex scatter [13]); cluster lensing confirms the $\eta \simeq 2.3$ residual independently of hydrostatic bias, so it is real, but it is the *same* residual, not a new one.

e. The morphology split (contested). The one lensing tension that does *not* reduce this way is the weak-lensing radial-acceleration split by morphology: early-type galaxies sit +0.26 dex above late types at fixed baryonic acceleration in the released KiDS catalogue [13], an 8.6–9.2σ contrast that is a_0- *and interpolation-independent* (a property-blind law predicts one relation; we confirmed the early − late model line cancels identically for every a_0 and ν). It was the framework's single largest tension. It is now *contested*: the radial-acceleration relation's own architects, re-analysing the same KiDS data with a stricter isolation cut, find the early/late deviation falls from $\sim 7.5\sigma$ to ~ 0 between $R_{\rm isol} = 2.7$ and 4.0 Mpc (the original used 3 Mpc), attributing it to early-type two-halo/satellite contamination plus a mild stellar-IMF offset [15]. Both ways: this relocates the split to a *selection* systematic rather than the intrinsic type-dependence the law forbids, but it rests on a smaller, noisier sample and does not erase the released-catalogue contrast. We therefore present it as an open selection-versus-intrinsic question, not as resolved; an *intrinsic* split, if it survives, falsifies the law.

f. Solar System (Cassini; PPN). Any covariant completion must respect the Cassini bound on the post-Newtonian parameter $|\gamma - 1| < 2.3\times10^{-5}$. The RAR demands a $\sim 30\%$ enhancement at the Solar radius that a naive AQUAL completion cannot reconcile with this bound; the fiducial tension is $\sim 8.7\sigma$, relieved to $\sim 1.9\sigma$ once the Galactic bulge contribution is removed [16]. It remains the first objection a referee raises at $z=0$, and the modified-inertia escape lacks a CMB-safe covariant form.

VII. THE ONE DISTINCTIVE PREDICTION: $a_0(z)$

Every confrontation above is *shared* with constant-a_0 modified dynamics: it tests the value of a_0, not its identification with Λ. The framework's only genuinely beyond-MOND content is a redshift dependence, because $a_0 \propto \sqrt{\rho_{\rm DE}}$. The prediction is conditional, and we state the condition sharply: if dark energy is a true cosmological constant, $\rho_{\rm DE}$ =const and $a_0(z) = a_0(0)$ is *flat*, exactly degenerate with constant-a_0 dynamics. The distinctive, declining branch

$$\frac{a_0(z)}{a_0(0)} = \sqrt{\frac{\rho_{\rm DE}(z)}{\rho_{\rm DE,0}}} \tag{7}$$

exists *only* if dark energy is dynamical—precisely the regime DESI's w_0w_aCDM preference ($w_0 > -1$, $w_a < 0$ at 2.8–4.2σ [17]) now indicates. The cleanest observable is the high-z zero-point of the BTFR: since $V^4 =$

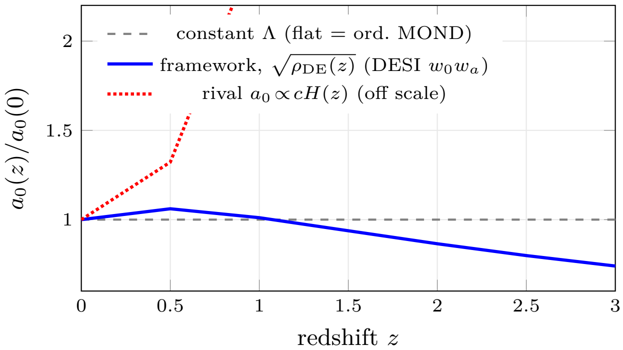

FIG. 2. The framework's distinctive prediction. On the DESI $w_0 w_a$ dark-energy history, $a_0(z) \propto \sqrt{\rho_{\rm DE}(z)}$ (blue) shows a $\sim 6\%$ bump near $z \simeq 0.5$ then declines to $0.74\, a_0(0)$ by $z = 3$; a true cosmological constant is flat (gray); the rival $a_0 \propto cH(z)$ (red) rises off scale. The most direct current datum, MUSE-DARK III [18], prefers the *rising* sign at $\sim 1.5\sigma$, against the framework's branch.

$GM_{\rm bar} a_0(z)$, a lower a_0 at high z shifts discs to lower V at fixed mass, $\delta \ln V = \frac{1}{4} \delta \ln a_0$, of order -0.03 dex ($\sim -7\%$ in velocity) by $z \simeq 3$ on the DESI branch—versus exactly zero for constant-a_0 dynamics. Figure 2 shows the three branches: on the DESI $w_0 w_a$ parametrization the framework's $a_0(z)$ rises by $\sim 6\%$ to a maximum near $z \simeq 0.5$ (a mild, distinctive feature) and then declines to $0.74\, a_0(0)$ by $z = 3$; a true cosmological constant gives a flat line; the rival reading $a_0 \propto cH(z)$ rises steeply off scale. These three are observationally distinct at $z \gtrsim 1$ given a ~ 0.01 dex BTFR zero-point measurement—$\mathcal{O}(10^3)$ discs with resolved kinematics and controlled gas fractions.

The interpolation function cancels in the ratio Eq. (7), so this prediction is interpolation-robust. The instruments are ELT/HARMONI, JWST/NIRSpec, and ALMA resolved high-z kinematics, cross-checked against DESI's $\rho_{\rm DE}(z)$; the framework predicts the two *track*, and a measured $a_0(z)$ that does not track DESI's $\rho_{\rm DE}(z)$ falsifies it. We note the $\sqrt{\rho_{\rm DE}}$ scaling is not itself novel (cf. early $a_0 \propto cH$ arguments); the fresh content is the specific DESI-history evaluation and the non-monotonic bump. *Honesty, both ways:* the most direct current measurement, MUSE-DARK III [18], fits a_0 *rising* with redshift—the wrong sign at face value—but the rise is degenerate with ΛCDM halo/baryon drift (reproduced in the Magneticum simulations with no fundamental a_0), and de-systematizing through that degeneracy collapses the residual to consistent with zero. The DESI dynamical-DE signal is sign-aligned but measures the *input* $\rho_{\rm DE}$, not a_0, so it cannot be cited as confirmation. The honest status is therefore *non-diagnostic at the systematic floor*: the direct datum leans against at face value but de-systematizes to null, the cosmological tailwind is uncashable, and the distinctive $z \gtrsim 1.5$ decline is untested below every current probe's floor. The single-redshift significance is moreover capped near 2–3σ by the DESI $w_0 w_a$ prior for any sample size; a clean 5σ needs a multi-redshift stack (e.g. $z = 3$ and $z = 5$) and an $\sim 2\times$ tighter dark-energy prior, realistically the 2030s.

VIII. RELATION TO OTHER EMERGENT-GRAVITY PROGRAMS

A QFT-trained reader will place this framework in the lineage of *emergent gravity*, where the Einstein equations arise as an equation of state from horizon thermodynamics [5, 6] and acceleration scales enter through the equipartition of horizon degrees of freedom. Verlinde's entropic-gravity proposal [7] similarly produces a MOND-like scale from the response of de Sitter entropy to baryonic matter and predicts an excess lensing tested (with mixed results) against KiDS. The present framework is more conservative than these in scope—it does not attempt to derive G or the Einstein equations—and more specific in one respect: it commits to the precise value $a_0 = c^2 \sqrt{\Lambda/32\pi}$ and to a single interpolation [Eq. (5)] across the isolated, external-field, and cosmological sectors, which makes it sharply falsifiable where the broader programs remain schematic. Its weakest point is shared with all of them: the $\mathcal{O}(1)$ coefficient relating a_0 to cH_Λ—here $1/Z$—is motivated by a horizon-area/equipartition counting but not forced, so the magnitude is a posit and only the redshift *scaling* Eq. (7) is a clean prediction. We regard honest engagement with that gap as more useful than a numerological derivation of Z.

IX. STANDING, FORECAST, AND OPEN PROBLEMS

Table I collects the falsifiable predictions with their instruments and the current lean. The honest summary is threefold. (1) The framework does not break ΛCDM and ΛCDM does not falsify it; there are no referee-proof kills in either direction. (2) Its $z = 0$ confrontations are shared with constant-a_0 modified dynamics, and it carries durable conceded losses—the cluster deficit, the CMB third-peak (where a_0 is absent from linear theory and a dust-mimic Ω must be inserted by hand, so the two-dark-sectors-as-one claim provably fails at recombination), Cassini, and the wrong-sign $a_0(z)$ lean. (3) Its genuinely distinctive content is Eq. (7), gated on dynamical dark energy, and it is sign-aligned with the one cosmological tension (DESI evolving DE) that points its way. The deepest open problem is theoretical: the coefficient $Z = \sqrt{32\pi/3}$ is an unforced posit. Until it is derived—candidate routes include the de Sitter–Unruh horizon thermodynamics that motivate Eq. (4), and the AeST free function—a_0 is fixed to Λ only up to an $\mathcal{O}(1)$ geometric factor, and the framework's predictive content lives entirely in the *ratio* Eq. (7), where Z cancels.

TABLE I. Falsifiable predictions, the instrument, and the current lean. "Distinctive" marks content beyond constant-a_0 modified dynamics.

Prediction	Instrument	Status
$a_0(z)\propto\sqrt{\rho_{\rm DE}}$ (distinct.)	DESI, ELT, JWST	non-diagnostic
high-z BTFR offset (distinct.)	JWST, ALMA	untested
EFE / SEP violation	Gaia, kinematics	contested 4σ
wide-binary boost +6.6%	Gaia DR4	contested
lensing RAR (α_∞)	Euclid, Rubin	soft pass
lensing = dynamics ($\Phi{=}\Psi$)	— (AeST identity)	reframes DM
cluster residual $\eta\simeq 2.3$	eROSITA, lens.	*fails*
morphology split $\to 0$	KiDS	*contested*
Cassini $\lvert\gamma{-}1\rvert$ bound	BepiColombo	1.9–8.7σ

X. CONCLUSION

For a reader who arrived knowing the equivalence principle, the Unruh effect, de Sitter thermodynamics, and DESI but not the radial acceleration relation: the claim on the table is that the galactic "dark matter" acceleration scale is the de Sitter temperature of the cosmological horizon, $a_0 = c^2\sqrt{\Lambda/32\pi}$, realized as modified inertia in the bath of Eq. (4); that this reproduces the radial acceleration and Tully–Fisher relations and predicts a strong-equivalence-principle violation that a particle dark sector forbids; that it inherits modified dynamics' real losses at clusters, in weak lensing, and in the Solar System, which we report at full weight; and that its one falsifiable distinctive signature is a dark-energy-tracking $a_0(z)$, now entering reach. The framework is, on the one axis that separates science from speculation, sharply falsifiable; on the axis that separates hypothesis from established physics, unproven. Both statements hold at once.

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Calculations herein—the eRASS1 cluster residual, the de Sitter–Unruh wide-binary boost, the Tolman-floor and density-law cluster tests, the interpolation-consistency audit, and the $a_0(z)$ forecast—were performed and cross-checked numerically; scripts and the both-ways audit ledgers are archived with this work.

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